

J. PARRY GERBER, PT, DSc, SCS, ATC¹ • ROBIN L. MARCUS, PT, PhD² • LELAND E. DIBBLE, PT, PhD, ATC³
 PATRICK E. GREIS, MD⁴ • ROBERT T. BURKS, MD⁴ • PAUL C. LASTAYO, PT, PhD, CHT⁵

Safety, Feasibility, and Efficacy of Negative Work Exercise Via Eccentric Muscle Activity Following Anterior Cruciate Ligament Reconstruction

Rehabilitation following anterior cruciate ligament reconstruction (ACL-R) has many components. A major area of focus, however, is on restoring muscle size, function, and performance, while preserving the newly re-established stability of the knee.^{4,6,13,49} Despite progressive and accelerated exercise programs, recalcitrant, long-term muscle impairments and functional deficits persist.^{2,16,18,26,42} Novel, more intensive rehabilitation protocols

following ACL-R are therefore needed to mitigate these muscle impairments and the related performance deficits. The safety and efficacy, however, of such protocols must first be established. Specifically, these postoperative rehabilitation programs must not induce any short or long-term injury to the healing tissues (primarily the ACL graft), surrounding soft tissue structures, and articular cartilage.^{13,14} This is especially important when considering an intervention designed to overcome muscle atrophy and weakness that utilizes high loads or induces high muscle forces. Resistance exercises biased toward eccentric (ECC) muscle contractions, where high loads create high muscle forces, may be ideally suited to reverse these muscle deficits following ACL-R. Because ECC contractions have historically been associated with injury and muscle damage and there has been no report of efficacy, there has been hesitancy to adopt postoperative programs biased toward ECC muscle activity despite the benefits that can be achieved from this type of exercise following ACL-R.

An acute exposure to high-force ECC muscle contractions can induce muscle

● **STUDY DESIGN:** Randomized, matched design.

● **BACKGROUND:** Optimal rehabilitation following anterior cruciate ligament reconstruction (ACL-R) requires safe and effective interventions. Negative work exercise (via eccentric muscle activity) has the potential to be highly effective at producing large quadriceps size and strength gains early after ACL-R. The purpose of this investigation was to evaluate the short-term safety and efficacy of adding a progressive negative work exercise program via eccentric (ECC) ergometry early after ACL-R.

● **METHODS AND MEASURES:** Beginning 3 weeks after ACL-R, 32 participants were randomly assigned into either a 12-week traditional (TRAD) or ECC exercise program. Safety was assessed by measuring knee pain, thigh pain, knee effusion, and knee stability prior to surgery and at 3, 15, and 26 weeks after surgery. Efficacy was assessed by measuring negative work output during the 12-week training program and by measuring functional ability (ie, quadriceps peak torque, hopping distance, self-reported functional ability and activity level scales) prior to surgery and 26 weeks after ACL-R.

● **RESULTS:** There were no significant differences between groups in measures of knee and thigh pain, effusion, or stability at any period after surgery. Negative work output increased systematically throughout training, while knee and thigh pain remained at relatively low levels. A significant group-by-time interaction was observed for quadriceps peak torque, hopping distance, and activity level ($P \leq .02$). Quadriceps strength and hopping distance of the involved limb improved by a significantly greater amount in the ECC group compared to the TRAD group ($P < .01$). Activity level decreased to a lesser extent in the ECC group compared to the TRAD group ($P = .02$).

● **CONCLUSIONS:** Negative work via an ECC intervention was implemented safely after ACL-R. The addition of negative work exercise also induced superior short-term results in strength, performance, and activity level after surgery. *J Orthop Sports Phys Ther* 2007;37(1):10-18. doi:10.2519/jospt.2007.2362

● **KEY WORDS:** ACL, knee, rehabilitation

¹ Doctoral candidate, Department of Exercise and Sport Science; Division of Physical Therapy, University of Utah, Salt Lake City, UT. ² Clinical Associate Professor, Division of Physical Therapy, University of Utah, Salt Lake City, UT; Clinical Associate Professor, Department of Exercise and Sport Science, University of Utah, Salt Lake City, UT. ³ Clinical Assistant Professor, Division of Physical Therapy, University of Utah, Salt Lake City, UT; Clinical Assistant Professor, Department of Exercise and Sport Science, University of Utah, Salt Lake City, UT. ⁴ Associate Professor, Department of Orthopedics, University of Utah, Salt Lake City, UT. ⁵ Associate Professor, Division of Physical Therapy, University of Utah, Salt Lake City, UT; Associate Professor, Department of Exercise and Sport Science, University of Utah, Salt Lake City, UT; Associate Professor, Department of Orthopedics, University of Utah, Salt Lake City, UT. The protocol for this research was approved by the University of Utah Institutional Review Board. Address correspondence to J. Parry Gerber, University of Utah Department of Exercise and Sport Science, 520 Wakara Way, Salt Lake City, UT 84108. E-mail: John.Gerber@amedd.army.mil

damage. However, a gradual and progressive exposure over time to these forces is a potent stimulus for increasing muscle size and strength.^{24,28,32,33} The greatest forces in muscle are produced when an external force exceeds that of the muscle and it lengthens eccentrically.³⁵ The resulting work performed by the muscle is termed “negative work” (when the muscle length change is opposite that of the muscle force vector, hence, work [force × distance] is negative). Improvements in quadriceps size and strength following negative work exercise have been superior to the changes following positive work exercise programs.^{28,32,33} Importantly, these improvements were achieved without harmful side effects such as undue joint or muscle pain. In support of these previous studies, we observed in a recent case report that progressive negative work induced impressive quadriceps size and strength gains following ACL-R.²⁴ The safety and efficacy of adding high-intensity negative work exercise to traditional (TRAD) rehabilitation early after ACL-R, however, has not fully been established.

Over the past 3 decades, many interventions, including those considered “accelerated” and of high intensity, have been reported as safe following ACL-R (ie, early range of motion [ROM], early weight-bearing, functional training, and weight-bearing exercises).^{9,25,39,44,45} Recently, Beynnon et al⁶ reported on rehabilitation programs that produced minimal strain on the ACL graft and compared that with one that produced increasingly greater strain on the graft. They found no differences in the anterior knee laxity, activity level, function, patient satisfaction, and response to cartilage metabolism biomarkers. They concluded that the intensity of these rehabilitation programs were safe, yet still emphasized caution with increasing the frequency and magnitude of quadriceps activity for concern of adversely increasing anterior knee lax-

ity or subjecting the healing knee to continued injury.⁶ Because the use of ECC exercise in the early postoperative period has been blunted due to concerns over implementing high external loads and muscle forces, we conducted a prospective, randomized clinical trial to evaluate the short-term safety and efficacy of adding a progressive negative work exercise program via ECC ergometry early after ACL-R. Our hypothesis was that adding a 12-week focused ECC resistance-training program 3 weeks after ACL-R would be equally as safe, and more efficacious, than current TRAD rehabilitation.

METHODS

POTENTIAL SUBJECTS WERE THOSE diagnosed with a rupture of the ACL at the University of Utah Sports Medicine Center between January 2004 and January 2005. Subjects were included in the study if they were between 18 and 50 years of age, moderately active prior to injury (Tegner⁴⁷ activity scale of 4 or higher) and willing to comply with postsurgical rehabilitation guidelines for 15 weeks following surgery. Subjects were excluded if they had a previous fracture or reconstructive surgery on either lower extremity, abnormal knee radiograph, concurrent injury to the posterior cruciate ligament or lateral collateral ligament, grade III tear of the medial collateral ligament, or a significant articular cartilage lesion observed during arthroscopy. Subjects with large vertical longitudinal meniscus tears were also excluded due to the concern of allowing aggressive rehabilitation in these cases. Two surgeons (P.E.G. and R.T.B.) performed all ACL-Rs for this study. An arthroscopically assisted autograft technique was used in all cases, using either the semitendinosus and gracilis in quadrupled fashion or the bone-patellar-tendon-bone (BTB) as a graft source. The graft selection was based on an educated and informed patient’s desire and/or request. The study received ap-

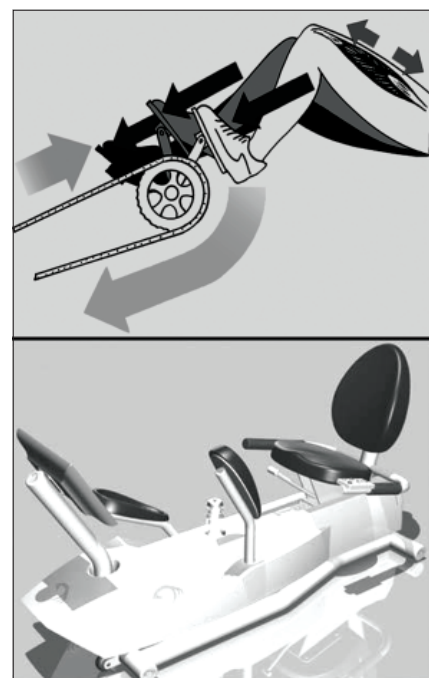


FIGURE 1. Two eccentric ergometers. As the pedals move towards the participant (to the right in these figures), he or she resists motor-propelled pedals from approximately 20° to 60° of knee flexion in a weight-bearing fashion. Because the magnitude of force produced by the motor exceeds that produced by the participant, the quadriceps work eccentrically, creating negative work.

proval from the Institutional Review Board at the University of Utah. All subjects signed informed-consent forms before participating, and the rights of all subjects were protected.

ECC and TRAD Rehabilitation Programs

Using a randomized matched design, subjects were randomly assigned after surgery into either an ECC or TRAD accelerated rehabilitation group. Subjects were matched by surgical procedure, sex, and age. For example, after the first 21-year-old female who had an ACL-R with a semitendinosus gracilis autograft was randomized (by coin flip) into 1 group, the second female having the same surgical procedure between the age of 18 and 24 was assigned to the other group.

All subjects followed a 3-phase accelerated rehabilitation program that was both criterion and time based (TABLE

RESEARCH REPORT

1).^{42,44} The exercise prescription was determined by the individual response to exercise. Specifically, if exercises were completed without an increase in knee pain or effusion, the subject was considered ready to progress. Other exercises were then added or current exercises were continued at a higher intensity, frequency, and/or duration.

After ligament reconstruction, all subjects completed 2 to 3 weeks of phase I exercises that focused on controlling pain and effusion, gaining full knee ROM, and attaining basic quadriceps function (TABLE 1). Beginning 3 weeks following ACL-R, subjects in the ECC group began a 12-week progressive negative work exercise program using 1 of 2 recumbent ECC ergometers, described previously in detail.^{24,33,34} The first ergometer was similar to a recumbent cycle. A second, more durable and comfortable ergometer (Eccentron, LLC, Denver, CO), similar to a recumbent stepper, was also used for eccentric training. The cycle ergometer was used for the first 5 subjects performing the ECC program and the stepper ergometer was used for the final 11 subjects. Both ergometers induced ECC contractions of the knee and hip extensors and negative work of similar magnitude (FIGURE 1). During each exercise session, the negative work rate was

visible on the computer monitor and the total amount of negative work (measured in kilojoules) was recorded. The pedal speed was self-selected and ranged from 20 to 40 revolutions per minute. Subjects were positioned on the ergometers for the negative work to occur from approximately 20° to 60° of knee flexion, effectively minimizing the possibility of a knee hyperextension injury. Intensity of exercise was based on the Borg Rating of Perceived Exertion Scale.³⁸ The first session was 5 minutes in duration at a “very, very light” intensity. Assuming a favorable individual response to exercise (ie, absence of increased knee pain, effusion, excessive fatigue, etc), subjects were allowed to gradually progress to a “hard” intensity and a maximum duration of 30 minutes (TABLE 2).

Beginning 3 weeks following surgery, subjects in the TRAD group continued with TRAD rehabilitation, and additionally, they were instructed to follow a similar exercise regime as the eccentric group, using a concentric ergometer (ie, gradually progress to a “hard” intensity and a duration of 30 minutes). All subjects were typically allowed 12 physical therapy visits and encouraged to complete physical therapy exercises as part of a home program. To assist with the home exercise program, all subjects had

access to resistance-training equipment, while only those in the ECC group had access to the ECC ergometer.

Tests and Measures

There are currently no standard measures to determine the safety and efficacy of an intervention following ACL-R, though many tests and measures have been used. In this study, a combination of clinical measures was used to establish safety: self-reported thigh and knee pain, joint effusion, and knee stability. Likewise, a combination of measures was used to establish efficacy: negative work output produced during the 12-week training program, strength and performance testing, and self-reported functional ability and activity level scales. All tests and measures were completed by the same investigator.

Safety

KNEE AND THIGH PAIN. During the 12-week training program, knee and thigh pain were assessed 3 times each week on nonconsecutive days (prior to exercise), using a 10-cm visual analog scale (VAS). Although it is common for patients to experience knee and thigh pain during rehabilitation following ACL-R, pain should generally decrease over time. The VAS was anchored at one end (0 cm)

TABLE 1

DETAILS OF THE POSTSURGICAL REHABILITATION PROGRAM*

POSTOPERATIVE PHASE	REHABILITATION REGIMEN
Phase 1. Duration, 2-3 wk	● Early range-of-motion exercises with an emphasis on gaining full knee extension ● Weight bearing as tolerated for the bone patellar-tendon bone procedure, touch weight bearing for the hamstring procedure ● Straight leg strengthening and gait training ● Goals for progression to phase 2: minimal pain and effusion, 0°-100° knee range of motion, good quadriceps contraction
Phase 2. Duration, 2-3 mo	● Endurance training (biking, stair stepper, etc) ● Progressive resistance training (leg press, calf press, mini-squats, hamstring curls, etc). An emphasis was placed initially on low resistance and multiple repetitions, and then gradually replaced with sets of increasing resistance and fewer repetitions ● Battery of balance exercises and beginning-level plyometric exercises ● Goals for progression to phase 3: full range of motion, hop on 1 leg without pain
Phase 3. Duration, 3-6 mo	● Continued progressive-resistance and endurance training ● Jogging/running progression and advanced plyometric exercises ● Advanced strengthening and functional exercise training to prepare the individual for full return to activity/sports ● Goals for returning to full activity: 90% strength and performance ability compared to the nonsurgical lower extremity

*For those in the eccentric group, the 12-week eccentric ergometry training program began 3 weeks after surgery.

TABLE 2

ECCENTRIC ERGOMETRY EXERCISE GUIDELINES, STARTING 3 WEEKS AFTER ANTERIOR CRUCIATE LIGAMENT RECONSTRUCTION

TRAINING WEEK	EXERCISE SESSION	INTENSITY (RPE)	DURATION (min)
1	1	Very, very light	5
	2	Very light	5-10
	3	Fairly light	5-10
2	4	Light	10-15
	5-6	Somewhat hard	10-15
3-4	7-12	Somewhat hard	15-20
5-6	13-18	Somewhat hard	20-25
7-8	19-24	Somewhat hard	26-30
9-12	25-36	Hard	26-30

Abbreviations: RPE, rating of perceived exertion.

Efficacy

NEGATIVE WORK OUTPUT. In general, an effective resistance-training program should result in systematic increases in a person's ability to perform the resistance exercise. In this study, negative work (kilojoules) for each ECC exercise session was recorded and averaged for each week. The ECC training program was considered efficacious if total work increased systematically throughout the training period.

QUADRICEPS STRENGTH. Quadriceps strength was assessed prior to surgery and 26 weeks after surgery using a Kin-Com isokinetic dynamometer (ChatteX, Chattanooga, TN). Isokinetic strength testing has been a common outcome measure after ACL-R, and good reliability has been established.^{8,40} Subjects were tested concentrically at $60^\circ \times s^{-1}$ in a seated position, with the hips and knees at 90° flexion and the thighs, pelvis, and upper body firmly strapped to the seat of the dynamometer. Prior to testing, a warm-up consisting of 3 repetitions at 50%, 75%, and 100% intensity was completed. After a brief rest period of 1 minute, subjects then completed 3 maximal repetitions at 100% intensity in an arc of 90° to 10° knee flexion. The peak torque of each repetition was averaged and used for statistical analysis.

PERFORMANCE. Subjects completed a single-leg long jump prior to surgery and 26 weeks after surgery, as a measure of performance. This test is 1 of the most widespread performance measures conducted after ACL-R and has been shown to have good reliability.^{8,43} After a 3- to 5-minute warm-up on a stationary bike, subjects were asked to stand on 1 leg, hop as far as possible, subsequently landing on the same leg. Arm position was self-selected, and swinging of the arms was permitted. Each leg was tested 3 times and maximal distance was recorded. The average of the 2 furthest hops was used for analysis.

SELF-REPORT. Subjective functional ability was assessed before training (3 weeks after surgery) and 26 weeks after surgery using the Activity of Daily Living Scale-Knee Outcome Survey (ADLS-KOS). The

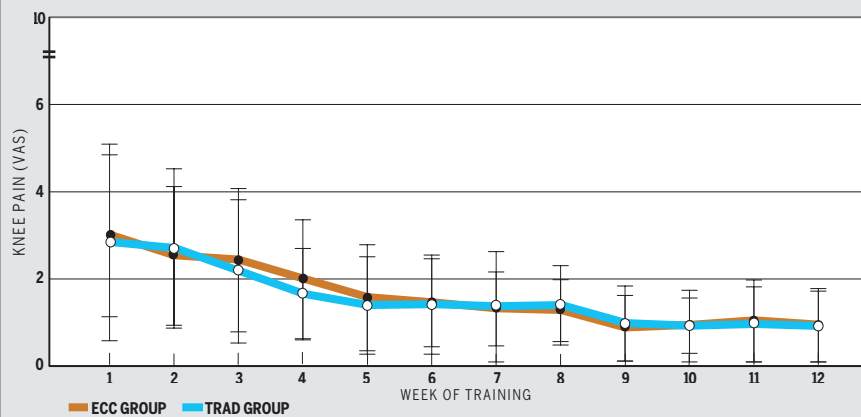


FIGURE 2. No significant differences in knee pain between the eccentric and traditional groups were observed at any time point during the 12 weeks of training ($P = .98$). Both the eccentric and traditional rehabilitation groups experienced an approximate 3-fold decrease in knee pain over the 12-week training period. (Error bars, 1 SD.)

by the descriptor “no pain” and at the other end (10 cm) by “extreme pain.” Weekly averages were utilized for statistical analysis.

KNEE EFFUSION. Effusion was assessed before training (3 weeks postoperatively), after training (15 weeks postoperatively), and during a follow-up examination (26 weeks postoperatively), by measuring joint circumference (mid patella). The difference in circumference between the surgical and nonsurgical knee 3 weeks after ACL-R was used as the baseline measure of effusion. A decrease in the surgical knee joint circumference 15 and 26 weeks after ACL-R would indicate a decrease in effusion and that rehabilitation was tolerated appropriately. The

intratester reliability was established for this variable via repeated circumference measurements of the nonsurgical knee during the same time periods ($ICC = 0.99$).

KNEE STABILITY. Stability of the postsurgical knee was assessed prior to surgery and 15 and 26 weeks after surgery, using the KT-1000 knee arthrometer, with anterior-posterior-directed manual maximal force. The KT-1000 is a commonly used instrument to evaluate knee stability and has good reliability and validity.^{8,11,22,41} The difference in laxity between the involved and uninvolved knees was recorded. Good intratester reliability ($ICC = 0.84$) was achieved using the KT-1000 in this study.

RESEARCH REPORT

ADLS-KOS is a 17-item questionnaire that measures symptoms and functional disability. This self-report instrument has demonstrated good reliability, validity, and responsiveness in assessing functional limitations relating to the knee.³⁰ Overall activity level (assessed by the Tegner Activity Scale⁴⁷) was completed prior to surgery and 26 weeks after surgery.

Statistical Analysis

A primary goal of the current study was to determine whether ECC and TRAD rehabilitation were equivalent (or equally safe). Because proving the null hypothesis is not statistically viable, we attempted to demonstrate that the ECC and TRAD rehabilitation groups did not differ substantially.⁷ Blackwelder⁷ suggested that failing to reject the null hypothesis with α set at .50 provides strong evidence of equivalence between groups. In addition, the use of confidence intervals (CI) has been recommended to improve interpretation of statistical results.^{1,7,48} A CI provides the range of values in which there is a 95% probability that the population parameter should reside. Interval estimates that overlap substantially around group means would indicate there is minimal difference between groups. Our approach of demonstrating equivalence was to report the means, 95% CIs, and P values for each safety measure. It would be reasonable to conclude that the ECC and TRAD groups were equally safe if the outcomes resulted in similar means, considerable overlap of the 95%

CIs, and P values greater than .50. In contrast, another goal of this study was to determine whether the addition of ECC resistance training was more efficacious than TRAD rehabilitation following ACL-R. For measures of efficacy, concluding that there was a difference between rehabilitation groups would be realized if the null hypothesis was rejected, with α set at .05.

This study was powered on the most objective safety measure employed in this study (ie, the KT-1000 assessment of knee stability). Hence, the sample size was based on the difference in laxity between the involved and uninvolved knees at the manual maximum anterior-posterior-directed force at 6 months post-ACL-R. A 2.0-mm side-to-side difference in knee laxity between the ECC and TRAD groups was considered practically significant. Using an online power calculator (<http://calculators.stat.ucla.edu/powercalc/>), a common standard deviation of 1.8 mm, 80% power, and α set at .05, a sample size of 14 subjects per group was estimated to be sufficient to detect a 2.0-mm difference between groups.

Data were analyzed with SPSS Version 13.0 (SPSS Inc, Chicago, IL). Descriptive statistics for categorical variables and measures of central tendency for continuous variables were calculated to summarize the data. Tests for outliers and assumptions of the parametric statistical tests were performed. An independent t test was used to analyze mean differences between the ECC and TRAD groups in knee laxity 15 and 26 weeks

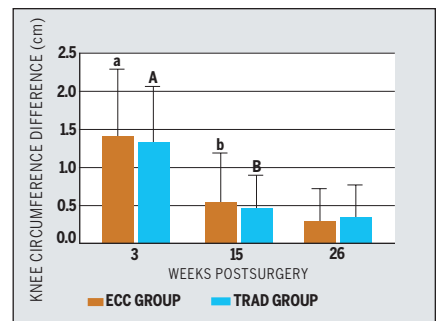


FIGURE 3. No significant differences between the eccentric and traditional groups were observed in the amount of effusion present in the surgical knee after anterior cruciate ligament reconstruction, as determined by mid-patella circumference measurements. (Error bars, 1 SD.) Significant decreases in effusion occurred in both the eccentric and traditional groups, between a and b and between A and B ($P < .001$).

after surgery. A 2-way mixed repeated-measure analysis of variance (ANOVA), with factors of group and time, was used to analyze mean differences between the ECC and TRAD rehabilitation groups for knee and thigh pain, effusion, quadriceps strength, hopping distance, ADLS-KOS, and activity level. These analyses included testing for a treatment effect, a time effect, and a treatment-by-time interaction effect.

RESULTS

NO SIGNIFICANT DIFFERENCES BETWEEN the ECC and TRAD rehabilitation groups in demographic characteristics of the 32 subjects (9 males and 7 females per group) were observed (P value range, .52-.90 [TABLE 3]). The semitendinosus gracilis graft was used to reconstruct the ACL in 5 males and 5 females in each group, while the BTB graft was used in 4 males and 2 females in each group. Because no statistically significant differences between surgical techniques were observed for any outcome measure, data were combined for analysis. The time between the initial ACL tear and surgery was longer than 1 year in 4 subjects (2 from each group). Excluding those cases, the time interval between initial injury and surgery was

TABLE 3

PRESURGICAL DEMOGRAPHIC INFORMATION (n = 16 IN EACH GROUP)

DESCRIPTION	ECCENTRIC GROUP MEAN (SD)	TRADITIONAL GROUP MEAN (SD)
Age (y)	29.4 (9.4)	31.0 (9.8)
Height (cm)	175.8 (8.9)	174.2 (8.5)
Body mass (kg)	78.4 (18.2)	76.4 (10.1)
ADLS-KOS	57.7 (7.8)	58.4 (7.9)
KT-1000 difference (mm)	5.6 (2.6)	6.1 (2.2)

Abbreviations: ADLS-KOS, Activity of Daily Living Scale-Knee Outcome Survey. No significant differences between the eccentric and traditional groups were observed for any above measure.

48.9 days (range, 23-116 days) in the ECC group and 71.5 days (range, 18-251 days) in the TRAD group. Two people in each group had meniscal lesions requiring repair, while 3 (ECC) and 2 (TRAD) had meniscal lesions requiring debridement and/or resection.

The 16 subjects in the ECC group completed 31.3 (range, 29-36) ergometry sessions for an overall 87% compliance rate. Overall, an average of 726 minutes of ECC ergometry was completed per participant. Subjects reported 15.8 days (via log, combined total during supervised rehabilitation and home exercise program) that lower extremity weight lifting was performed, excluding ECC ergometry, during the 12-week training period. In comparison, the 16 subjects in the TRAD group averaged 689 minutes of concentric ergometry. Subjects reported 25.1 days that lower extremity weight lifting was performed during the 12-week training period. The number of reported weight lifting days was significantly greater in the TRAD rehabilitation group ($P < .01$).

Safety

KNEE AND THIGH PAIN. Knee pain in the ECC group (mean, 2.9; 95% CI, 1.78 to 4.34) and the TRAD group (mean, 2.8; 95% CI, 1.42 to 4.23) did not differ before training ($P = .91$). No significant differences in knee pain between the ECC and TRAD groups were observed at any time point during the 12 weeks of training ($P = .98$). There was also no significant difference in knee pain 15 weeks after surgery between the ECC (mean, 0.77; 95% CI, 0.19 to 1.35) and the TRAD (mean, 0.86; 95% CI, 0.25 to 1.48) groups ($P = .95$). Both groups experienced a significant decrease in knee pain from the third to fifteenth week post-surgery (ECC, $P < .01$; TRAD, $P < .02$). Average knee pain at the beginning of training for both groups was approximately 3 (on the 10-point VAS) and decreased gradually to below 1 by the ninth week of training (FIGURE 2).

Thigh pain in the ECC group (mean, 0.86; 95% CI, 0 to 3.71) and the TRAD

TABLE 4	POSTSURGICAL KT-1000 ARTHROMETER MEASUREMENTS AT THE MANUAL MAXIMUM FORCE*			
	15 wk		26 wk	
MANUAL MAXIMUM	MEAN (SD)	95% CI	MEAN (SD)	95% CI
Eccentric group	1.9 (1.5)	1.1 to 2.7	2.1 (1.5)	1.3 to 2.9
Traditional group	1.6 (0.9)	1.1 to 2.1	2.2 (1.2)	1.5 to 3.0

Abbreviations: CI, confidence interval.
*Values denote laxity differences (millimeters) in absolute value between involved and uninvolved knees. No significant knee laxity differences were observed between the eccentric and traditional groups (P value range, .54-.78).

group (mean, 0.78; 95% CI, 0 to 3.2) did not differ before training ($P = .42$). No significant differences in thigh pain between the ECC and TRAD groups were observed at any time point during the 12 weeks of training ($P = .57$). There was also no significant difference in thigh pain 15 weeks after surgery between the ECC (mean, 1.42; 95% CI, 0 to 2.94) and the TRAD (mean, 1.33; 95% CI, 0 to 2.72) groups ($P = .48$).

KNEE EFFUSION. Effusion in the ECC group (mean, 1.41 cm; 95% CI, 0.98 to 1.83 cm) and the TRAD group (mean, 1.31 cm; 95% CI, 0.89 to 1.74 cm) did not differ before training ($P = .75$). There was no significant difference in knee effusion 15 weeks after surgery between the ECC (mean, 0.53 cm; 95% CI, 0.22 to 0.84 cm) and the TRAD (mean, 0.41 cm; 95% CI, 0.10 to 0.71 cm) groups ($P = .72$). There was no significant difference in knee effusion 26 weeks after surgery between the ECC (mean, 0.31 cm; 95% CI, 0.13 to 0.49 cm) and the TRAD (mean, 0.34 cm; 95% CI, 0.16 to 0.53 cm) groups ($P = .72$). Both groups experienced a significant decrease in effusion from the third to fifteenth week postsurgery ($P < .001$). The amount of effusion decreased at each subsequent period (FIGURE 3).

KNEE STABILITY. There were no significant anterior-posterior knee laxity differences between groups at 15 or 26 weeks after ACL-R using the KT-1000 at the manual maximum directed force (P value range, .54-.78). The 95% CI was within 2 mm (the value deemed

clinically significant) during measurements at both periods (TABLE 4).

Efficacy

NEGATIVE WORK OUTPUT. After the 2-week "ramp-up" phase of ECC ergometry (TABLE 2), values of negative work increased over 3-fold by completion of the training program. Negative work values increased progressively each of the 12 weeks of training while knee and thigh pain remained at relatively low levels (FIGURE 4).

QUADRICEPS STRENGTH. A significant group-by-time interaction was observed for quadriceps strength (peak torque) of the involved thigh ($P < .04$). Compared to preoperative testing, quadriceps peak torque of the involved thigh 26 weeks after surgery significantly increased in the ECC group ($P < .01$), but not in the TRAD group ($P = .74$). Mean quadriceps peak torque in the ECC group was 124.2 Nm (95% CI, 104.3 to 144.0 Nm) preoperatively and 159.2 Nm (95% CI, 137.0 to 181.4 Nm) 26 weeks after surgery. Mean peak torque in the TRAD group was 125.1 Nm (95% CI, 105.2 to 163.1 Nm) preoperatively and 128.9 Nm (95% CI, 95.0 to 162.6 Nm) 26 weeks after surgery (FIGURE 5).

PERFORMANCE. A significant group-by-time interaction was observed for hopping distance of the involved lower extremity ($P < .01$). Compared to preoperative testing, hopping distance of the involved lower extremity 26 weeks after surgery significantly increased in the ECC group ($P < .01$) but not in the TRAD group ($P = .17$). Mean hopping

RESEARCH REPORT

distance in the ECC group was 80.3 cm (95% CI, 62.9 to 97.7 cm) preoperatively and 112.9 cm (95% CI, 94.0 to 131.8 cm) 26 weeks after surgery. Mean hopping distance in the TRAD group was 87.9 cm (95% CI, 71.4 to 104.4 cm) preoperatively and 96.2 cm (95% CI, 78.3 to 114.1 cm) 26 weeks after surgery (FIGURE 5).

SELF-REPORT. No significant group-by-time interaction was observed for scores on the ADLS-KOS ($P = .66$). Both groups experienced a significant increase in ADLS-KOS scores from 3 weeks (before training) to 26 weeks postsurgery ($P < .01$). Scores in the ECC group improved from 43.3 (95% CI, 38.0 to 48.6) before training to 71.7 (95% CI, 67.9 to 75.5) 26 weeks after surgery, and scores in the TRAD group improved from 45.4 (95% CI, 39.1 to 51.8) before training to 70.4 (95% CI, 65.8 to 75.1) 26 weeks after surgery.

A significant group-by-time interaction was observed for activity level (based on the Tegner Activity Scale) in this study ($P = .02$). Activity level, from preinjury to 26 weeks after surgery, decreased to a lesser extent in the ECC group compared to the TRAD group. Mean activity level decreased in the ECC group from 6.7 (95% CI, 6.0 to 7.4) preinjury to 5.4 (95% CI, 4.6 to 6.2) 26 weeks postsurgery. Mean activity level decreased in the TRAD group from 7.0 (95% CI, 6.2 to 7.8) preinjury to 4.6 (95% CI, 3.7 to 5.5) 26 weeks postsurgery.

DISCUSSION

THE RESULTS OF THIS STUDY SUPPORT our hypotheses, that adding a 12-week focused ECC resistance-training program 3 weeks after ACL-R would be equally as safe, and more efficacious, than current TRAD rehabilitation. The addition of negative work exercise induced superior short-term results in strength, performance, and activity level after surgery. Recovery of quadriceps strength and hopping ability was amplified and returned more quickly with ECC training. The gradual and progressive exposure to

negative work augmented muscle function following ACL-R, while being equally as safe as TRAD rehabilitation (ie, no differences in knee or thigh pain, effusion, or knee stability).

The rehabilitation program used for the experimental group in this study was biased toward ECC muscle contractions, which are known to produce high muscle forces. A primary benefit of high-force resistance training is the greater potential to overload muscles, which in turn produces greater increases in muscle size and strength.^{28,32,33} Our results support previous studies as strength was significantly greater with the addition of negative work via ECC ergometry after ACL-R. A secondary benefit of negative work exercise is the potential for an enhancing effect on performance-related tasks. The single-leg long jump test, for example, is a functional measure of strength, confidence, and dynamic muscle coactivation.³¹ A successful hop test after ACL-R depends largely on the person's ability and confidence to control the landing or ECC phase of the hop. Intuitively, an ECC intervention would seem ideally suited to improve such a task, which is exactly what was observed in this study.

These positive results do not suggest that aggressive ECC interventions should be implemented without caution after ACL-R. Despite the benefits, ECC contractions have been associated with muscle damage in individuals not adapted to these forces. The subjects in this study achieved high negative work levels, but due to the gradual, progressive, and individualized nature of the program, they were able to tolerate the high forces of ECC resistance exercise without any adverse responses. The careful titration of these forces is important, because muscle tissue, articular cartilage, tendon, and ligament adapt favorably to controlled overload by structurally adapting to stresses placed upon them.^{14,37} While ECC muscle contractions are potentially damaging to those naive to eccentrically biased exercise, repeated and progressive exposure to the eccentric activity is protective.^{10,15}

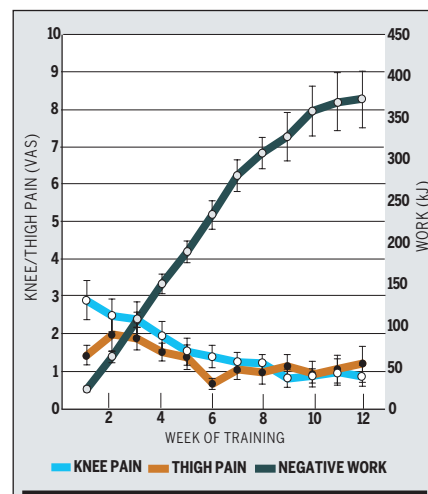


FIGURE 4. Both knee and thigh pain remained at low levels in individuals who trained eccentrically while negative work dramatically increased.

In this study, we safeguarded against potential harmful effects of the ECC intervention by allowing individuals to be gradually exposed to longer durations and greater external loads, which results in higher force production in muscle, only if they had no increase in knee pain, effusion or a reduction in their quadriceps-force-producing abilities. In the current study, it was not uncommon for subjects to experience knee discomfort or quadriceps soreness at some time during the training program. Decreasing the negative work, by lowering the exertion level and/or the duration of exercise (and in some instances canceling an exercise session), typically resolved these symptoms. The overall success of adding negative work exercise in this study was likely contingent on these principles (ie, gradual exercise progression, progressive muscle overload, and carefully monitoring the individual response to exercise).

Another important safety consideration was the manner in which ECC training was performed. The ECC intervention was a weight-bearing exercise with resistance applied during 20° to 60° of knee flexion. Exercising within these parameters avoided increases in joint pressure and knee pain compared to other exercise alternatives (ie, non-weight-bearing or weight-bearing exercises outside of

those ROM constraints).^{9,17,20,46} Stress on the healing ACL graft is also less during weight-bearing exercises from 20° to 60° flexion. Extensive work on the strain behavior of the ACL during various quadriceps-dominated, weight-bearing exercises, which include both concentric and eccentric components (ie, squatting, concentric ergometer, lunges, and step-up and step-down maneuvers), has demonstrated that ACL strain was greatest near terminal knee extension and less between 20° to 60° flexion.^{4,6,21,27} Perhaps the most relevant aspect of these studies is that increasing the resistance during weight-bearing exercise did not increase ACL strain, while conversely, increasing the resistance during non-weight-bearing exercise increased ACL strain significantly. Strains placed on the ACL during weight-bearing exercises were attenuated when higher-resistance torques were used.^{4,5,21,23,27} The results of our study (no differences in knee pain, effusion, ligament laxity) add support to these previous reports. The clinical application of these findings suggests that adding a quadriceps intervention that induces high muscle forces within certain parameters (weight-bearing and limited ROM) can be tolerated and safely implemented following ACL-R.

Limitations

Several limitations characterize the current study. Although general rehabilitation guidelines were provided, consider-

able variation in exercise prescription and exercise progression could have occurred between the treating therapists. In addition, while attempts were made to equilibrate total exercise time between rehabilitation groups, those in the ECC group had more overall exposure to resistance training. Although this increase in ECC training was found safe after ACL-R, it is unclear whether the ECC mode of training or simply an increased total exposure to resistance training was responsible for the positive efficacy results observed in this study. Future studies are needed to clarify this issue.

While pain, effusion, and decreased function may indicate that an exercise intervention is too aggressive to be considered safe, the absence of these findings is not necessarily equivalent to joint safety. Others have used more advanced technological instruments as measures of safety after ACL-R, like enzyme-linked assays to examine cartilage biomarkers or technetium scintiscans to assess joint metabolic activity.^{6,14} Although these advances provide very useful information, they are unfortunately more costly and not easily accessible. In contrast, the measures employed in this study are clinically applicable and practical.

An additional limitation of this study is that we cannot determine the long-term safety and effectiveness of ECC training after ACL-R for the 2 different graft types (ie, hamstring or the BTB autograft). In this study we analyzed combined results of the 2 graft types, because no statistically significant differences between groups were observed for any outcome measure. However, power could have been insufficient to detect differences between the graft types, due to the small sample size and individual variability. The purpose of this study was limited to determining the safety of eccentric exercise for those undergoing ACL-R independent of surgery type and to begin to understand whether this mode of resistance exercise is efficacious during the early stages of rehabilitation.

Increased degenerative changes of the

knee have been shown to occur after ACL-R.^{12,36} It is currently unknown whether the high forces related to ECC training early after ACL-R would accelerate these changes. Studies using the enzyme-linked assays or technetium scintiscans may be a way to investigate this in the future. Incidentally, degenerative changes after ACL-R have also been associated with decreased quadriceps strength.^{3,19,29} If ECC training leads to increased quadriceps size and strength early following ACL-R, and if these muscle improvements persist, it is conceivable that this type of intervention could actually provide a long-term protective effect to the knee. Further research would be required to support this notion.

CONCLUSION

THIS STUDY DEMONSTRATES THAT negative work via an ECC intervention can be implemented safely 3 weeks after ACL-R. The addition of negative work exercise also induced superior short-term results in strength, performance, and activity level after surgery. With the increased understanding of short-term safety and efficacy provided by this study, further research should be conducted to corroborate these findings and to focus on long-term benefits of ECC interventions following ACL-R. ●

REFERENCES

1. American Psychological Association. *Publication Manual of the American Psychological Association*. 5th ed. Washington, DC: American Psychological Association; 2001.
2. Arangio GA, Chen C, Kalady M, Reed JF, 3rd. Thigh muscle size and strength after anterior cruciate ligament reconstruction and rehabilitation. *J Orthop Sports Phys Ther*. 1997;26:238-243.
3. Baker KR, Xu L, Zhang Y, et al. Quadriceps weakness and its relationship to tibiofemoral and patellofemoral knee osteoarthritis in Chinese: the Beijing osteoarthritis study. *Arthritis Rheum*. 2004;50:1815-1821.
4. Beynnon BD, Fleming BC, Johnson RJ, Nichols CE, Renstrom PA, Pope MH. Anterior cruciate ligament strain behavior during rehabilitation exercises in vivo. *Am J Sports Med*. 1995;23:24-34.

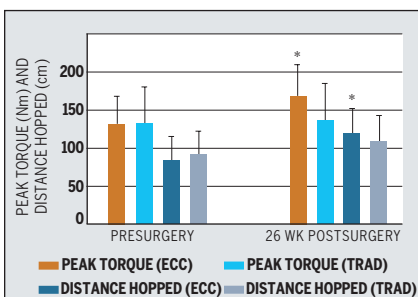


FIGURE 5. Quadriceps peak torque and distance hopped of the involved lower extremity before surgery and 26 weeks after surgery. *Quadriceps peak torque and hopping distance increased significantly from pre-surgery to 26 weeks post surgery in the ECC group ($P<.01$). (Error bars = 1SD)

5. Beynnon BD, Johnson RJ, Fleming BC, Stankewich CJ, Renstrom PA, Nichols CE. The strain behavior of the anterior cruciate ligament during squatting and active flexion-extension. A comparison of an open and a closed kinetic chain exercise. *Am J Sports Med.* 1997;25:823-829.
6. Beynnon BD, Uh BS, Johnson RJ, et al. Rehabilitation after anterior cruciate ligament reconstruction: a prospective, randomized, double-blind comparison of programs administered over 2 different time intervals. *Am J Sports Med.* 2005;33:347-359.
7. Blackwelder WC. "Proving the null hypothesis" in clinical trials. *Control Clin Trials.* 1982;3:345-353.
8. Brosky JA, Jr., Nitz AJ, Malone TR, Caborn DN, Rayens MK. Intrarater reliability of selected clinical outcome measures following anterior cruciate ligament reconstruction. *J Orthop Sports Phys Ther.* 1999;29:39-48.
9. Bynum EB, Barrack RL, Alexander AH. Open versus closed chain kinetic exercises after anterior cruciate ligament reconstruction. A prospective randomized study. *Am J Sports Med.* 1995;23:401-406.
10. Clarkson PM, Hubal MJ. Exercise-induced muscle damage in humans. *Am J Phys Med Rehabil.* 2002;81:S52-69.
11. Daniel DM, Stone ML, Dobson BE, Fithian DC, Rossman DJ, Kaufman KR. Fate of the ACL-injured patient. A prospective outcome study. *Am J Sports Med.* 1994;22:632-644.
12. Daniel DM, Stone ML, Sachs R, Malcom L. Instrumented measurement of anterior knee laxity in patients with acute anterior cruciate ligament disruption. *Am J Sports Med.* 1985;13:401-407.
13. Dye SF. The future of anterior cruciate ligament restoration. *Clin Orthop Relat Res.* 1996;130-139.
14. Dye SF, Wojtys EM, Fu FH, Fithian DC, Gillquist I. Factors contributing to function of the knee joint after injury or reconstruction of the anterior cruciate ligament. *Instr Course Lect.* 1999; 48:185-198.
15. Ebbeling CB, Clarkson PM. Exercise-induced muscle damage and adaptation. *Sports Med.* 1989;7:207-234.
16. Ejerhed L, Kartus J, Sernert N, Kohler K, Karlsson J. Patellar tendon or semitendinosus tendon autografts for anterior cruciate ligament reconstruction? A prospective randomized study with a two-year follow-up. *Am J Sports Med.* 2003;31:19-25.
17. Escamilla RF, Fleisig GS, Zheng N, Barrentine SW, Wilk KE, Andrews JR. Biomechanics of the knee during closed kinetic chain and open kinetic chain exercises. *Med Sci Sports Exerc.* 1998; 30:556-569.
18. Feller JA, Webster KE. A randomized comparison of patellar tendon and hamstring tendon anterior cruciate ligament reconstruction. *Am J Sports Med.* 2003;31:564-573.
19. Fisher NM, Pendergast DR. Reduced muscle function in patients with osteoarthritis. *Scand J Rehabil Med.* 1997;29:213-221.
20. Fitzgerald GK. Open versus closed kinetic chain exercise: issues in rehabilitation after anterior cruciate ligament reconstructive surgery. *Phys Ther.* 1997;77:1747-1754.
21. Fleming BC, Beynnon BD, Renstrom PA, Peura GD, Nichols CE, Johnson RJ. The strain behavior of the anterior cruciate ligament during bicycling. An in vivo study. *Am J Sports Med.* 1998; 26:109-118.
22. Fleming BC, Brattbakk B, Peura GD, Badger GJ, Beynnon BD. Measurement of anterior-posterior knee laxity: a comparison of three techniques. *J Orthop Res.* 2002;20:421-426.
23. Fleming BC, Ohlen G, Renstrom PA, Peura GD, Beynnon BD, Badger GJ. The effects of compressive load and knee joint torque on peak anterior cruciate ligament strains. *Am J Sports Med.* 2003;31:701-707.
24. Gerber JP, Marcus RL, Dibble LE, Greis PE, LaStayo PC. Early application of negative work via eccentric ergometry following anterior cruciate ligament reconstruction: a case report. *J Orthop Sports Phys Ther.* 2006;36:298-307.
25. Haggmark T, Eriksson E. Cylinder or mobile cast brace after knee ligament surgery. A clinical analysis and morphologic and enzymatic studies of changes in the quadriceps muscle. *Am J Sports Med.* 1979;7:48-56.
26. Harilainen A, Sandelin J, Jansson KA. Cross-pin femoral fixation versus metal interference screw fixation in anterior cruciate ligament reconstruction with hamstring tendons: results of a controlled prospective randomized study with 2-year follow-up. *Arthroscopy.* 2005;21:25-33.
27. Heijne A, Fleming BC, Renstrom PA, Peura GD, Beynnon BD, Werner S. Strain on the anterior cruciate ligament during closed kinetic chain exercises. *Med Sci Sports Exerc.* 2004; 36:935-941.
28. Hortobagyi T, Dempsey L, Fraser D, et al. Changes in muscle strength, muscle fibre size and myofibrillar gene expression after immobilization and retraining in humans. *J Physiol.* 2000;524 Pt 1:293-304.
29. Hurley MV. Quadriceps weakness in osteoarthritis. *Curr Opin Rheumatol.* 1998;10:246-250.
30. Irrgang JJ, Snyder-Mackler L, Wainner RS, Fu FH, Harner CD. Development of a patient-reported measure of function of the knee. *J Bone Joint Surg Am.* 1998;80:1132-1145.
31. Keays SL, Bullock-Saxton J, Keays AC. Strength and function before and after anterior cruciate ligament reconstruction. *Clin Orthop Relat Res.* 2000;174-183.
32. LaStayo PC, Ewy GA, Pierotti DD, Johns RK, Lindstedt S. The positive effects of negative work: increased muscle strength and decreased fall risk in a frail elderly population. *J Gerontol A Biol Sci Med Sci.* 2003;58:M419-424.
33. LaStayo PC, Pierotti DJ, Pifer J, Hoppeler H, Lindstedt SL. Eccentric ergometry: increases in locomotor muscle size and strength at low training intensities. *Am J Physiol Regul Integr Comp Physiol.* 2000;278:R1282-1288.
34. Lastayo PC, Reich TE, Urquhart M, Hoppeler H, Lindstedt SL. Chronic eccentric exercise: improvements in muscle strength can occur with little demand for oxygen. *Am J Physiol.* 1999; 276:R611-615.
35. Lindstedt SL, LaStayo PC, Reich TE. When active muscles lengthen: properties and consequences of eccentric contractions. *News Physiol Sci.* 2001;16:256-261.
36. Lohmander LS, Ostengren A, Englund M, Roos H. High prevalence of knee osteoarthritis, pain, and functional limitations in female soccer players twelve years after anterior cruciate ligament injury. *Arthritis Rheum.* 2004;50:3145-3152.
37. Muneta T, Yamamoto H, Takakuda K, Sakai H, Furuya K. Effects of postoperative immobilization on the reconstructed anterior cruciate ligament. An experimental study in rabbits. *Am J Sports Med.* 1993;21:305-313.
38. Noble BJ, Borg GA, Jacobs I, Ceci R, Kaiser P. A category-ratio perceived exertion scale: relationship to blood and muscle lactates and heart rate. *Med Sci Sports Exerc.* 1983; 15:523-528.
39. Noyes FR, Mangine RE, Barber S. Early knee motion after open and arthroscopic anterior cruciate ligament reconstruction. *Am J Sports Med.* 1987;15:149-160.
40. Pincivero DM, Lephart SM, Karunakara RA. Reliability and precision of isokinetic strength and muscular endurance for the quadriceps and hamstrings. *Int J Sports Med.* 1997;18:113-117.
41. Robnett NJ, Riddle DL, Kues JM. Intertester reliability of measurements obtained with the KT-1000 on patients with reconstructed anterior cruciate ligaments. *J Orthop Sports Phys Ther.* 1995;21:113-119.
42. Rosenberg TD, Franklin JL, Baldwin GN, Nelson KA. Extensor mechanism function after patellar tendon graft harvest for anterior cruciate ligament reconstruction. *Am J Sports Med.* 1992;20:519-525; discussion 525-516.
43. Ross MD, Langford B, Whelan PJ. Test-retest reliability of 4 single-leg horizontal hop tests. *J Strength Cond Res.* 2002;16:617-622.
44. Shelbourne KD, Nitz P. Accelerated rehabilitation after anterior cruciate ligament reconstruction. *Am J Sports Med.* 1990;18:292-299.
45. Shelbourne KD, Trumper RV. Preventing anterior knee pain after anterior cruciate ligament reconstruction. *Am J Sports Med.* 1997;25:41-47.
46. Steinkamp LA, Dillingham MF, Markel MD, Hill JA, Kaufman KR. Biomechanical considerations in patellofemoral joint rehabilitation. *Am J Sports Med.* 1993;21:438-444.
47. Tegner Y, Lysholm J. Rating systems in the evaluation of knee ligament injuries. *Clin Orthop Relat Res.* 1985;43-49.
48. Westlake WJ. Statistical problems in interpreting comparative bioavailability trials. *Int J Clin Pharmacol Biopharm.* 1975;11:342-345.
49. Wilk KE. Are there speed limits in rehabilitation? *J Orthop Sports Phys Ther.* 2005;35:50-51.